

13_objectCreation_wGenes

August 31, 2026

```
[ ]: library(Seurat)
# library(SeuratDisk)

library(reticulate)
library(anndata)

library(ggplot2)
library(ggpubr)
library(pheatmap)
library(dplyr)
library(tidyr)
library(RColorBrewer)
#library(clustree)

#library(UpSetR)
library(grid)

#library(PRRDC)
library(Matrix)

getwd()

dataset_id <- "simulated_mm_RA_wGenes"
genome_id <- "mm10"
samples <- c("all")#, "old", "young")

for (sample in samples) {
  dir.create(paste0("figures_", dataset_id, "_", sample))
}

colorTools <- c( # "MATES"="#F98A7B",
  "STARsolo_TE" = "#FFBC81",
  "STARsolo_TE_EM" = "#f0df93",
  "SoloTE_unique" = "#A4DD9B",
  "SoloTE_thr2" = "#6FC69D",
  "SoloTE_thr1" = "#4BB2BB",
  "SoloTE_thr0" = "#3989BF",
```

```
"Stellarscope" = "#4F5D93",  
"simulated" = "grey70"  
)  
  
thrMinCells <- 500 * 0.05
```

Loading required package: SeuratObject

Loading required package: sp

Attaching package: 'SeuratObject'

The following objects are masked from 'package:base':

intersect, t

Attaching package: 'anndata'

The following object is masked from 'package:SeuratObject':

Layers

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

```
expand, pack, unpack
```

```
'/mnt/transfer/TEbenchmarking/Rscripts'
```

```
Warning message in dir.create(paste0("figures_", dataset_id, "_", sample)):  
"'figures_simulated_mm_RA_wGenes_all' already exists"
```

```
[ ]: # import TE annotation with conversion columns for SoloTE and Stellarscope  
conversion_table <- read.table(paste0("annotation/annotation_", genome_id, "  
  ↪"_conversion_withAge.tsv"))  
conversion_table$soloteID <- gsub("\\\\?", "", conversion_table$soloteID)
```

```
[ ]: library(rtracklayer)  
  
gtf <- rtracklayer::import("annotation/gencode.vM10.primary_assembly.annotation.  
  ↪gtf") # annotation used for mapping in snakemake workflow  
gtf <- as.data.frame(gtf)  
ensembl_conversion <- gtf[,c('gene_id', 'gene_name')]  
ensembl_conversion <- unique(ensembl_conversion)
```

```
Loading required package: GenomicRanges
```

```
Loading required package: stats4
```

```
Loading required package: BiocGenerics
```

```
Attaching package: 'BiocGenerics'
```

```
The following objects are masked from 'package:dplyr':
```

```
combine, intersect, setdiff, union
```

```
The following object is masked from 'package:SeuratObject':
```

```
intersect
```

```
The following objects are masked from 'package:stats':
```

```
IQR, mad, sd, var, xtabs
```

```
The following objects are masked from 'package:base':
```

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
Position, rank, rbind, Reduce, rownames, sapply, setdiff, sort,
table, tapply, union, unique, unsplit, which.max, which.min

Loading required package: S4Vectors

Attaching package: 'S4Vectors'

The following objects are masked from 'package:Matrix':

expand, unname

The following object is masked from 'package:tidyr':

expand

The following objects are masked from 'package:dplyr':

first, rename

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Attaching package: 'IRanges'

The following objects are masked from 'package:dplyr':

```
collapse, desc, slice
```

The following object is masked from ‘package:sp’:

```
%over%
```

Loading required package: GenomeInfoDb

Create list of objects with counts generated by each colorTools

```
[5]: objList <- list()

objGeneList <- list()
```

1 Splatter simulation

```
[6]: splatter_objs <- list()
splatter_objs_genes <- list()
samples <- c("all")

main_path <- "/mnt/transfer2/minnow_simulate_mm_RA_TE_"

for (sample in samples) {
  print(sample)

  path <- paste0(main_path, sample, "_TEsNgenes/alevin/")

  splatter_mat <- read.table(paste0(path, "quants_mat.csv"),
                             sep = ",", header = FALSE)

  # remove NAs, which are present because rows end with ","
  splatter_mat <- splatter_mat %>% select_if(~ !all(is.na(.)))
  colnames(splatter_mat) <- read.table(paste0(path, "quants_mat_cols.txt"))$V1
  rownames(splatter_mat) <- read.table(paste0(path, "quants_mat_rows.txt"))$V1
  # transpose matrix
  splatter_mat <- t(splatter_mat)

  print("loci in conversion table:")
  print(table(rownames(splatter_mat) %in% conversion_table$locusID))

  splatter_mat_TEs <- splatter_mat[rownames(splatter_mat) %in% ↵
↵conversion_table$locusID,] # keep only TEs
```

```

    rownames(splatter_mat_TEs) <- gsub(pattern = "_", replacement = "-",
↪ rownames(splatter_mat_TEs)) # convert underscores in the TE names to match
↪ the annotation

    splatter_mat_genes <- splatter_mat[!rownames(splatter_mat) %in%
↪ conversion_table$locusID,] # everything except TEs that are in the
↪ conversion table (which means only the genes)
    #rownames(splatter_mat_genes) <- sub("\\.|\\d+$", "",
↪ rownames(splatter_mat_genes)) # remove version number
    # convert to ENSEMBL
    rownames(splatter_mat_genes) <-
↪ ensembl_conversion$gene_name[match(rownames(splatter_mat_genes), ensembl_conversion$gene_id,
↪ )]
    rownames(splatter_mat_genes) <- make.unique(rownames(splatter_mat_genes),
↪ sep = "_")

    splatter_objs[[sample]] <- Seurat::CreateSeuratObject(splatter_mat_TEs,
    project = paste0("simulated_", sample),
    min.cells = thrMinCells, min.features = 0
    )
    print(splatter_objs[[sample]])
    # around 100 features are removed because expressed in less than the
↪ threshold of cells
    print("loci matching stellarscope ID:")
    print(table(rownames(splatter_objs[[sample]]) %in%
↪ conversion_table$stellarscopeID))

    print("Summary of n counts")
    print(summary(splatter_objs[[sample]]$nCount_RNA))
    print("Summary of n features")
    print(summary(splatter_objs[[sample]]$nFeature_RNA))

    splatter_objs_genes[[sample]] <- Seurat::
↪ CreateSeuratObject(splatter_mat_genes,
    project = paste0("simulated_", sample),
    min.cells = thrMinCells, min.features = 0
    )
    print(splatter_objs_genes[[sample]])

    print("Summary of n counts")
    print(summary(splatter_objs_genes[[sample]]$nCount_RNA))
    print("Summary of n features")
    print(summary(splatter_objs_genes[[sample]]$nFeature_RNA))
}

```

```
[1] "all"
[1] "loci in conversion table:"

FALSE TRUE
1986 5273

Warning message:
"Data is of class matrix. Coercing to dgCMatrix."

An object of class Seurat
4858 features across 500 samples within 1 assay
Active assay: RNA (4858 features, 0 variable features)
1 layer present: counts
[1] "loci matching stellarscope ID:"
```

```
TRUE
4858
[1] "Summary of n counts"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
20960  37978   43850   44391   49869   80386
[1] "Summary of n features"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 2437   3052   3194   3181   3318   3711
```

```
Warning message:
"Feature names cannot have underscores ('_'), replacing with dashes ('-')"
Warning message:
"Data is of class matrix. Coercing to dgCMatrix."
```

```
An object of class Seurat
1831 features across 500 samples within 1 assay
Active assay: RNA (1831 features, 0 variable features)
1 layer present: counts
[1] "Summary of n counts"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 7822  13887   16155   16328   18292   30417
[1] "Summary of n features"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   941   1151   1196   1194   1243   1419
```

1.1 Normalize splatter

```
[7]: # TEs
for (sample in samples) {
  splatter_objs[[sample]] <- NormalizeData(splatter_objs[[sample]],
    normalization.method = "LogNormalize",
    scale.factor = 10000
  )
}
```

Normalizing layer: counts

```
[8]: # Genes
for (sample in samples) {
  splatter_objs_genes[[sample]] <-
  ↪ NormalizeData(splatter_objs_genes[[sample]],
    normalization.method = "LogNormalize",
    scale.factor = 10000
  )
}
```

Normalizing layer: counts

2 STARsolo TE

```
[24]: STARsolo_TE_objs <- list()
STARsolo_TE_objs_genes <- list()

main_path <- paste0("/mnt/transfer/results/STARsolo_EM_TEwGenes/", dataset_id,
  ↪ "/" )

for (sample in samples) {
  print(sample)

  path <- paste0(main_path, sample, "/TE_Solo.out/Gene/raw/")

  filteredBarcodes <- read.table(paste0(main_path, sample, "/TE_Solo.out/Gene/
  ↪ filtered/barcodes.tsv"))$V1

  STARsolo_TE_mat <- Seurat::ReadMtx(
    mtx = paste0(path, "matrix.mtx"),
    cells = paste0(path, "barcodes.tsv"),
    features = paste0(path, "features.tsv")
  ) # read matrix
  STARsolo_TE_mat <- STARsolo_TE_mat[, filteredBarcodes] # keep only barcodes
  ↪ passing thresholds

  # split TEs and genes
  STARsolo_TE_mat_genes <- STARsolo_TE_mat[rownames(STARsolo_TE_mat) %in%
  ↪ ensembl_conversion$gene_name,]
  STARsolo_TE_mat_TEs <- STARsolo_TE_mat[rownames(STARsolo_TE_mat) %in%
  ↪ conversion_table$locusID,]

  # create gene object
```



```

STARsolo_TE_objs_genes[[sample]] <- Seurat::
↪CreateSeuratObject(STARsolo_TE_mat_genes,
  project = paste0("STARsolo_TE_", sample),
  min.cells = thrMinCells, min.features = 0
)
print("Summary of n gene counts")
print(summary(STARsolo_TE_objs_genes[[sample]]$nCount_RNA))
print("Summary of n gene features")
print(summary(STARsolo_TE_objs_genes[[sample]]$nFeature_RNA))

# create TE object
STARsolo_TE_objs[[sample]] <- Seurat::
↪CreateSeuratObject(STARsolo_TE_mat_TEs,
  project = paste0("STARsolo_TE_", sample),
  min.cells = thrMinCells, min.features = 0
)

# Remove some classes of TEs (already not present in SoloTE)
classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")
starsoloTEs <- Features(STARsolo_TE_objs[[sample]])
filteredStarsoloTEs <- conversion_table[conversion_table$stellarscopeID_
↪%in% starsoloTEs, ] %>%
  dplyr::filter(!class %in% classesToExclude) %>%
  pull(stellarscopeID)
print("Percentage of TEs in removed orders:")
print(length(setdiff(starsoloTEs, filteredStarsoloTEs)) /_
↪length(starsoloTEs) * 1000)
STARsolo_TE_objs[[sample]] <-_
↪STARsolo_TE_objs[[sample]][intersect(starsoloTEs, filteredStarsoloTEs), ]

print("Summary of n TE counts")
print(summary(STARsolo_TE_objs[[sample]]$nCount_RNA))
print("Summary of n TE features")
print(summary(STARsolo_TE_objs[[sample]]$nFeature_RNA))

print("Loci present in the annotation:")
print(table(rownames(STARsolo_TE_objs[[sample]])) %in%_
↪conversion_table$stellarscopeID))
}

# add to the list of objects
objList[["STARsolo_TE"]] <- STARsolo_TE_objs
objGeneList[["STARsolo_TE"]] <- STARsolo_TE_objs_genes

```

```

[1] "all"
[1] "Summary of n gene counts"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

      6801    11813    13628    13796    15468    25120
[1] "Summary of n gene features"
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
      831    1020    1060    1056    1099    1256

Warning message:
"Feature names cannot have underscores ('_'), replacing with dashes ('-')"

[1] "Percentage of TEs in removed orders:"
[1] 0.08934155
[1] "Summary of n TE counts"
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
    15401    26005    29588    29876    33319    51462
[1] "Summary of n TE features"
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
     2813     3245     3334     3331     3423     3765
[1] "Loci present in the annotation:"

TRUE
11192

```

3 STARsolo TE with EM

```

[26]: STARsolo_TE_EM_objs <- list()
      STARsolo_TE_EM_objs_genes <- list()

main_path <- paste0("/mnt/transfer/results/STARsolo_EM_TeWGenes/", dataset_id,
  ↪"/")

for (sample in samples) {
  print(sample)

  path <- paste0(main_path, sample, "/TE_Solo.out/Gene/raw/")
  filteredBarcodes <- read.table(paste0(main_path, sample, "/TE_Solo.out/Gene/
  ↪filtered/barcodes.tsv"))$V1

  STARsolo_TE_EM_mat <- Seurat::ReadMtx(
    mtx = paste0(path, "UniqueAndMult-EM.mtx"),
    cells = paste0(path, "barcodes.tsv"),
    features = paste0(path, "features.tsv")
  ) # read matrix
  STARsolo_TE_EM_mat <- STARsolo_TE_EM_mat[, filteredBarcodes] # keep only
  ↪barcodes passing thresholds

  # split TEs and genes
  STARsolo_TE_EM_mat_genes <- STARsolo_TE_EM_mat[rownames(STARsolo_TE_EM_mat),
  ↪%in% ensembl_conversion$gene_name,]

```

```

STARsolo_TE_EM_mat_TEs <- STARsolo_TE_EM_mat[rownames(STARsolo_TE_EM_mat),
↳%in% conversion_table$locusID,]

# create gene object
STARsolo_TE_EM_objs_genes[[sample]] <- Seurat::
↳CreateSeuratObject(STARsolo_TE_EM_mat_genes,
  project = paste0("STARsolo_TE_EM_", sample),
  min.cells = thrMinCells, min.features = 0
)
print("Summary of n gene counts")
print(summary(STARsolo_TE_objs_genes[[sample]]$nCount_RNA))
print("Summary of n gene features")
print(summary(STARsolo_TE_objs_genes[[sample]]$nFeature_RNA))

# create TE object
STARsolo_TE_EM_objs[[sample]] <- Seurat::
↳CreateSeuratObject(STARsolo_TE_EM_mat_TEs,
  project = paste0("STARsolo_TE_EM_", sample),
  min.cells = thrMinCells, min.features = 0
)
# Remove some classes of TEs (already not present in SoloTE)
classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")
starsoloTEs <- Features(STARsolo_TE_EM_objs[[sample]])
filteredStarsoloTEs <- conversion_table[conversion_table$stellarscopeID,
↳%in% starsoloTEs, ] %>%
  dplyr::filter(!class %in% classesToExclude) %>%
  pull(stellarscopeID)
print("Percentage of TEs in removed orders:")
print(length(setdiff(starsoloTEs, filteredStarsoloTEs)) /
↳length(starsoloTEs) * 100)
STARsolo_TE_EM_objs[[sample]] <-
↳STARsolo_TE_EM_objs[[sample]][intersect(starsoloTEs, filteredStarsoloTEs), ]

print("Summary of n counts")
print(summary(STARsolo_TE_EM_objs[[sample]]$nCount_RNA))
print("Summary of n feature")
print(summary(STARsolo_TE_EM_objs[[sample]]$nFeature_RNA))

print("Loci present in the annotation:")
print(table(rownames(STARsolo_TE_EM_objs[[sample]])) %in%
↳conversion_table$stellarscopeID))
}

# add to the list of objects
objList[["STARsolo_TE_EM"]] <- STARsolo_TE_EM_objs
objGeneList[["STARsolo_TE_EM"]] <- STARsolo_TE_EM_objs_genes

```

```

[1] "all"
[1] "Summary of n gene counts"
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
6801   11813   13628   13796   15468   25120
[1] "Summary of n gene features"
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
831      1020    1060    1056    1099    1256

Warning message:
"Feature names cannot have underscores ('_'), replacing with dashes ('-')"

[1] "Percentage of TEs in removed orders:"
[1] 0.04551183
[1] "Summary of n counts"
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
21918   36304   41082   41453   45910   70122
[1] "Summary of n feature"
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
5216     6201    6430    6422    6669    7474
[1] "Loci present in the annotation:"

TRUE
28551

```

4 SoloTE Default (thr 255)

```

[27]: SoloTE_TEobjs <- list()
SoloTE_Gene_objs <- list()

main_path <- paste0("/mnt/transfer/results/SoloTEout/", dataset_id, "/")

for (sample in samples) {
  print(sample)

  path <- paste0(main_path, sample, "/", sample, "_SoloTE_output/", sample,
↳ "_legacytes_MATRIX/")
  # read matrix
  SoloTE_mat <- Seurat::ReadMtx(
    mtx = paste0(path, "matrix.mtx"),
    cells = paste0(path, "barcodes.tsv"),
    features = paste0(path, "features.tsv")
  )
  # filter cells present in the simulation
  filteredBarcodes <- colnames(splatter_objs[[sample]])
  SoloTE_mat <- SoloTE_mat[,filteredBarcodes]

```

```

# select locus level TEs
TEs <- grep("SoloTE", rownames(SoloTE_mat), value = T)
locusTEs <- grep("chr", TEs, value = T)
# select genes
genes <- setdiff(rownames(SoloTE_mat), TEs)

# Genes
# subset the matrix keeping only genes
SoloTE_GENEmat <- SoloTE_mat[genes, ]

# create Seurat object with shallow filtering of genes expressed in at
↳ least 50 cells and cells expressing at least 100 genes
SoloTE_Gene_objs[[sample]] <- Seurat::CreateSeuratObject( SoloTE_GENEmat,
  project = "SoloTE_unique_",
  min.cells = thrMinCells, min.features = 0
)

print("Summary of n counts")
print(summary(SoloTE_Gene_objs[[sample]]$nCount_RNA))
print("Summary of n features")
print(summary(SoloTE_Gene_objs[[sample]]$nFeature_RNA))

# TEs
# subset the matrix keeping only TEs
SoloTE_TEmat <- SoloTE_mat[locusTEs, ]
# remove "SoloTE" from the name of the TEs and substitute "/" and "_" with
↳ "-"
rownames(SoloTE_TEmat) <- gsub("SoloTE\\|", "", rownames(SoloTE_TEmat))
rownames(SoloTE_TEmat) <- gsub("\\|", "-", rownames(SoloTE_TEmat))
rownames(SoloTE_TEmat) <- gsub("\\_", "-", rownames(SoloTE_TEmat))
rownames(SoloTE_TEmat) <- gsub("\\?", "", rownames(SoloTE_TEmat))

table(rownames(SoloTE_TEmat) %in% conversion_table$soloteID)
# transform into Stellarscope IDs
rownames(SoloTE_TEmat) <-
↳ conversion_table$stellarscopeID[match(rownames(SoloTE_TEmat),
↳ conversion_table$soloteID)]

# create Seurat object with shallow filtering of TEs expressed in at least
↳ thrMinCells cells and cells expressing at least 20 TEs
SoloTE_TEobjs[[sample]] <- Seurat::CreateSeuratObject( SoloTE_TEmat,
  project = paste0("SoloTE_unique_", sample),
  min.cells = thrMinCells, min.features = 0
)

print("Summary of n counts")
print(summary(SoloTE_TEobjs[[sample]]$nCount_RNA))

```

```

print("Summary of n features")
print(summary(SoloTE_TEobjs[[sample]]$nFeature_RNA))

print("Loci present in the annotation:")
print(table(rownames(SoloTE_TEobjs[[sample]]) %in%
↪conversion_table$stellarscopeID))
}

objList[["SoloTE_unique"]] <- SoloTE_TEobjs
objGeneList[["SoloTE_unique"]] <- SoloTE_Gene_objs

```

```

[1] "all"
[1] "Summary of n counts"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
7168  12204   14044   14144   15822   24988
[1] "Summary of n features"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  908    1096    1136    1134    1180    1329
[1] "Summary of n counts"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
15596  25055   28308   28492   31627   47531
[1] "Summary of n features"
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 1893    2302    2388    2382    2470    2718
[1] "Loci present in the annotation:"

TRUE
5491

```

5 SoloTE thr 2,1 and 0

```

[30]: SoloTE_TEobjs <- list()
SoloTE_Gene_objs <- list()

for(thr in c("thr2","thr1","thr0")){

  main_path <- paste0("/mnt/transfer/results/SoloTEout_",thr,"/", dataset_id,
↪"/")

  for (sample in samples){
    print(sample)

    path <- paste0(main_path, sample, "/", sample, "_SoloTE_output/",
↪sample, "_legacytes_MATRIX/")
    # read matrix
    SoloTE_mat <- Seurat::ReadMtx(

```

```

    mtx = paste0(path, "matrix.mtx"),
    cells = paste0(path, "barcodes.tsv"),
    features = paste0(path, "features.tsv")
)
# filter cells
filteredBarcodes <- colnames(splatter_objs[[sample]])

SoloTE_mat <- SoloTE_mat[,filteredBarcodes]
# select locus level TEs
TEs <- grep("SoloTE", rownames(SoloTE_mat), value = T)
locusTEs <- grep("chr", TEs, value = T)
# select genes
genes <- setdiff(rownames(SoloTE_mat), TEs)

# subset the matrix keeping only genes
SoloTE_GENEmat <- SoloTE_mat[genes, ]
# create Seurat object with shallow filtering of genes expressed in at
↳ least 50 cells and cells expressing at least 100 genes
SoloTE_Gene_objs[[sample]] <- Seurat::CreateSeuratObject(SoloTE_GENEmat,
  project = paste0("SoloTE_",thr,"_", sample),
  min.cells = thrMinCells, min.features = 0
)

# subset the matrix keeping only TEs
SoloTE_TEmat <- SoloTE_mat[locusTEs, ]
# remove "SoloTE" from the name of the TEs and substitute "/" and "-"
↳ with "-"
rownames(SoloTE_TEmat) <- gsub("SoloTE\\|", "", rownames(SoloTE_TEmat))
rownames(SoloTE_TEmat) <- gsub("\\|", "-", rownames(SoloTE_TEmat))
rownames(SoloTE_TEmat) <- gsub("\\_", "-", rownames(SoloTE_TEmat))
rownames(SoloTE_TEmat) <- gsub("\\?", "", rownames(SoloTE_TEmat))

table(rownames(SoloTE_TEmat) %in% conversion_table$soloteID)
# transform into Stellarscope IDs
rownames(SoloTE_TEmat) <-
↳ conversion_table$stellarscopeID[match(rownames(SoloTE_TEmat),
↳ conversion_table$soloteID)]

# create Seurat object with shallow filtering of TEs expressed in at
↳ least thrMinCells cells and cells expressing at least 20 TEs
SoloTE_TEobjs[[sample]] <- Seurat::CreateSeuratObject(SoloTE_TEmat,
  project = paste0("SoloTE_",thr,"_", sample),
  min.cells = thrMinCells, min.features = 0
)

summary(SoloTE_TEobjs[[sample]]$nCount_RNA)

```

```

summary(SoloTE_TEobjs[[sample]]$nFeature_RNA)

print("Loci present in the annotation:")
print(table(rownames(SoloTE_TEobjs[[sample]]) %in%
↪conversion_table$stellarscopeID))
}

objList[[paste0("SoloTE_",thr)]] <- SoloTE_TEobjs
objGeneList[[paste0("SoloTE_",thr)]] <- SoloTE_Gene_objs
}

```

```

[1] "all"
[1] "Loci present in the annotation:"

```

```

TRUE
7343
[1] "all"
[1] "Loci present in the annotation:"

```

```

TRUE
9604
[1] "all"
[1] "Loci present in the annotation:"

```

```

TRUE
26663

```

6 Stellarscope

```

[33]: Stellarscope_objs <- list()

main_path <- paste0("/mnt/transfer/results/stellarscope_out/", dataset_id,
↪"_pseudobulk_onestep/")

for (sample in samples){
  print(sample)
  path <- paste0(main_path, sample, "/")
  # read matrix
  Stellarscope_mat <- Seurat::ReadMtx(mtx = paste0(path, sample,
↪"_pseudobulk-TE_counts.mtx"),
                                cells = paste0(path,
↪sample, "_pseudobulk-barcodes.tsv"),
                                features = paste0(path,
↪sample, "_pseudobulk-features.tsv"),
                                feature.column = 1) # read matrix
  # remove the unassigned counts

```



```

no_feature <- Stellarscope_mat["__no_feature",]
Stellarscope_mat <-
↪Stellarscope_mat[setdiff(rownames(Stellarscope_mat), "__no_feature"),]
  # substitute "_" with "-"
  rownames(Stellarscope_mat) <- gsub(pattern = "_", replacement = "-",
↪rownames(Stellarscope_mat))

# filter cells passing filters
filteredBarcodes <- colnames(splatter_objs[[sample]])
Stellarscope_mat <- Stellarscope_mat[,filteredBarcodes]

# create Seurat object with shallow filtering of TEs expressed in at least
↪5% of cells and cells expressing at least 20 TEs
Stellarscope_objs[[sample]] <- Seurat::CreateSeuratObject(Stellarscope_mat,
↪project = paste0("Stellarscope_", sample),
  min.cells = thrMinCells, min.features = 0)

# Remove some classes of TEs (already not present in SoloTE)
classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")
stellarscopeTEs <- Features(Stellarscope_objs[[sample]])

filteredStellarscopeTEs <- conversion_table[conversion_table$stellarscopeID,
↪%in% stellarscopeTEs,] %>%
  filter(! class %in% classesToExclude) %>% pull(stellarscopeID)
  print("percentage of loci removed")
  print(length(setdiff(stellarscopeTEs, filteredStellarscopeTEs))/
↪length(stellarscopeTEs) * 100) # percentage removed

Stellarscope_objs[[sample]] <-
↪Stellarscope_objs[[sample]][intersect(stellarscopeTEs,
↪filteredStellarscopeTEs),]

print("n count summary")
print(summary(Stellarscope_objs[[sample]]$nCount_RNA))
print("n features summary")
print(summary(Stellarscope_objs[[sample]]$nFeature_RNA))

print("Loci present in the annotation:")
print(table(rownames(Stellarscope_objs[[sample]]) %in%
↪conversion_table$stellarscopeID))
}

objList[["Stellarscope"]] <- Stellarscope_objs

```

```

[1] "all"
[1] "percentage of loci removed"
[1] 0.1383673

```

```
[1] "n count summary"
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
24704   38148   42463   42767   46914   70135
[1] "n features summary"
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 4524    5088    5195    5191    5308    5832
[1] "Loci present in the annotation:"

TRUE
15156
```

6.1 Normalize objects

```
[34]: for (sample in samples){
      for (tool in names(objList)){
        objList[[tool]][[sample]] <- NormalizeData(objList[[tool]][[sample]],
          normalization.method = "LogNormalize", scale.factor = 10000)
      }
    }
```

Normalizing layer: counts

Normalizing layer: counts

Normalizing layer: counts

Normalizing layer: counts

Normalizing layer: counts

Normalizing layer: counts

Normalizing layer: counts

7 Save objects

```
[ ]: dir.create("workspaces")
      save.image("workspaces/13_evaluation_objectCreation_wGenes.Rdata")
```

Warning message in dir.create("workspaces"):
 "'workspaces' already exists"

```
[ ]:
```